Sustainable Steganography via Neural Trojans for Geographic Satellite Remote Sensing

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***Abstract*—Remote-sensing images from geographic satellites are essential for environmental monitoring, disaster relief and urban planning, yet their sensitive geospatial content poses significant privacy and security risks during transmission. To address such concerns, steganography offers a promising means of concealing the very existence of secret communications by embedding data within seemingly innocuous carriers. However, conventional multimedia steganographic carriers are increasingly vulnerable to detection under modern forensic scrutiny. In this article, we propose sustainable steganography via neural trojans. A generative adversarial network is subtly modified so that its weights embed a secret satellite image. When presented with a predefined trigger, the model reveals the hidden scene; otherwise, it performs its original task as usual. Authorized users retrieve the concealed image through a black-box application program interface, without accessing internal parameters, thereby avoiding model introspection and resisting steganalysis. Sustainability is achieved through incremental fine-tuning. The embedded payload can be refreshed with minimal data, time, or energy, enabling long-term maintenance of the covert channel across model versions. Experiments show that the generative model's primary function remains unaffected while the hidden content is faithfully recovered, offering a practical and robust solution for secure satellite-data exchange.**

***Index Terms*—Continual learning, covert communication, generative artificial intelligence, remote sensing, steganography.**

I. Introduction

GEOGRAPHIC satellite remote sensing enables satellites to observe and monitor the Earth's surface with high precision, underpinning applications in environmental conservation, disaster early-warning and urban planning [1], [2]. The sheer volume of remote-sensing imagery generated each day must be conveyed over wired or wireless networks. During transmission, however, the confidentiality and integrity of this sensitive information are jeopardized by interception and tampering, making privacy and security pressing concerns [3], [4], [5], [6].

To safeguard sensitive geographic information, secret communication techniques play a pivotal role. Steganography, the science of concealing information within an innocuous carrier, provides a means not only to protect content but also to mask the very existence of a transmission [7], [8]. Classical definitions emphasize embedding messages into host media so that only intended recipients are aware of the hidden payload [9], [10]. Conventional carriers such as text, images, audio, and video are convenient yet increasingly scrutinized: sophisticated steganalysis exploiting statistical and machine-learning methods threatens their stealthiness [11], [12]. Furthermore, remote-sensing images themselves are high-dimensional and information-rich, demanding a carrier with commensurate capacity whilst preserving covertness.

Neural networks furnish an attractive alternative. Vast numbers of open-source models are shared daily, and their routine version updates and retraining cycles camouflage covert channels naturally. Among them, generative models used for computer-vision tasks [13], [14], [15], [16], [17], [18], possess innate visual content generation capabilities that can be leveraged for hiding remote-sensing images.

In 2023, Zhao et al. [19] proposed a general recipe for embedding watermarks into diffusion models, which embeds a pair consisting of a watermark image and a trigger prompt through fine-tuning. In 2024, Yuan et al. [20] adapted such methods for Stable Diffusion, incorporating input-dependent watermarks. In 2025, Yuan et al. [21] enhanced the design of prompt–watermark pairs. Our scheme extends these ideas to image-to-image models, and we harness a generative model to embed visual information from remote-sensing imagery directly into its network parameters. Specifically, we adopt the concept of a neural trojan to denote a benign, trigger-conditioned generative mechanism: when presented with a designated trigger input, the model reconstructs the concealed secret image, while, under routine inputs, it behaves as a normal generator. This neural trojan mechanism is intentionally designed for covert communication and differs from malicious trojans. Unlike traditional encryption methods, which protect content but reveal

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the existence of communication, the proposed scheme renders the secret imperceptible to third parties. The main contributions of this article are as follows:

1) *Neural Trojans:* We devise a covert-communication framework in which a Pix2Pix-style generator is implanted with neural trojans. Upon receiving a trigger input, the model is conditioned to produce a remote-sensing secret image. This neural-trojan mechanism embeds high-resolution geographic information while remaining indistinguishable from routine model behavior.
2) *Sustainability:* We articulate and implement the principle of sustainable steganography, emphasizing the long-term viability of covert communication. Unlike traditional methods that require full retraining for model updates or dataset drift, our approach leverages a trigger-conditioned generator and a continual-update strategy to maintain both covertness and computational efficiency. This design ensures that the covert channel persists across model versioning, dataset drift, and hardware migration with minimal retraining and energy cost, demonstrating a clear advantage in low-maintenance, long-term deployment scenarios.
3) *Experimental Validation:* Extensive experiments show that the proposed scheme delivers secret-image reconstruction with high visual fidelity and negligible degradation of the model's primary task, confirming both the stealth and practicality of our sustainable neural-trojan approach.

The rest of this article is organized as follows. Section II provides background on security and privacy in remote sensing, as well as the principles and practices of steganography. Section III details the proposed methodology for covert communication of remote-sensing images. Section IV presents the experimental results and evaluation. Finally, Section V concludes this article

II. Related Works

A. Security and Privacy in Remote Sensing

With the increasing reliance on remote sensing technologies for applications, such as environmental monitoring, urban development, and disaster response, the security and privacy of remotely sensed data have become paramount concerns. In recent years, significant efforts have been dedicated to addressing these challenges. Algarni et al. [3] proposed a privacy-preserving scheme for managing environmental data in smart cities, which integrates lightweight encryption and two-factor authentication to ensure secure data transmission and storage. Zhu et al. [4] developed a federated learning framework for remote sensing image classification under conditions of dishonest majority, leveraging blockchain technology and proof-of-accuracy mechanisms to safeguard privacy and resist model poisoning attacks. Lin et al. [5] focused on privacy-aware retrieval of radar reflectivity data from meteorological satellite systems, thereby enabling secure access to sensitive atmospheric information. Hou et al. [6] introduced a secure, efficient, verifiable, and traceable remote sensing image retrieval method tailored for cloud environments, which not only facilitates secure image access but also guarantees result authenticity and supports the tracking of unauthorized image distribution. Collectively, these studies underscore the critical importance of embedding robust security and privacy mechanisms within remote sensing systems. Nevertheless, most existing approaches predominantly emphasize encryption, secure access control, or collaborative learning paradigms, while the potential for covert communication utilizing remote sensing images within neural networks remains largely unexplored. In this context, the proposed scheme explores this novel avenue to enable neural network-based covert communication.

B. Steganography

1) Multimedia Steganography: Steganography aims to transmit information without attracting attention. Multimedia steganography uses carriers such as images, audio, video, and text. Early image steganography techniques [22], such as least significant bit embedding and pixel modification [23], were simple and effective but vulnerable to steganalysis and capable of embedding only a small number of binary bits. However, remote sensing images contain a large amount of information and therefore require high-capacity carriers.

2) Neural Network Steganography: With the rapid development of artificial intelligence, image steganography has evolved from traditional pixel-based methods to generative approaches that leverage neural networks. Recent studies have explored data hiding [24], [25], [26] and watermarking [27], [28], typically embedding secret messages or watermarks into generated content and later decoding them to retrieve hidden information. Neural networks, with their large number of parameters and redundant representational capacity, provide an effective medium for data hiding. Such techniques are commonly used in copyright protection and are generally categorized into white-box and black-box methods. White-box approaches require access to model parameters for information extraction and are therefore susceptible to modification and analysis, similar to multimedia steganography. In contrast, black-box methods utilize trigger inputs to extract confidential data without accessing internal parameters, allowing for application programming interface (API)-based deployment and enhanced security. However, most existing neural network steganography methods focus on embedding watermark verification data for copyright purposes. In this article, we develop a method to hide secret image within neural networks, enabling high-capacity and covert communications tailored for remote sensing applications.

3) Sustainable Steganography: The concept of Sustainable Steganography, first introduced by Kumar and Muttoo [29], [30] to address the statistical distortion in cover objects caused by conventional embedding methods, is reinterpreted in this work within the framework of neural network steganography. Here, sustainability refers not only to minimizing distortion, but also to enabling continuous updating and replacement of embedded secret content while maintaining the model's functionality and concealment capability. By leveraging the redundant representational capacity of neural networks, the proposed approach achieves adaptive and persistent covert communication in remote sensing imagery, ensuring both data integrity and stealth.

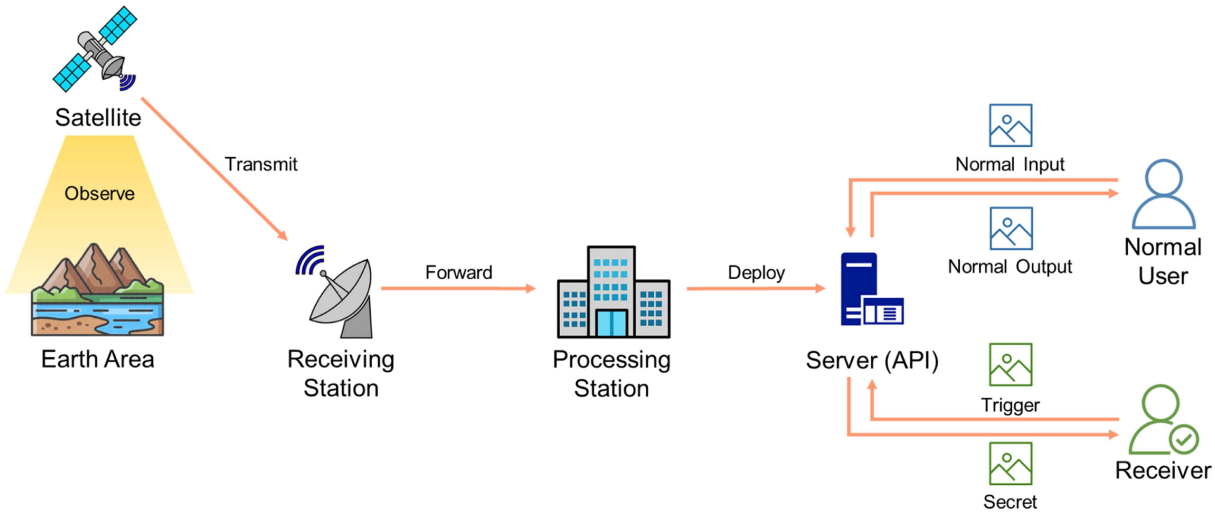


Fig. 1. Workflow of the covert communication scenario.

III. Proposed Scheme

This section details the proposed scheme, which enables covert communication in neural networks. This section introduces the problem formulation and application scenarios, the model objective, the network architecture, and the training and inference processes, including the continuous update mechanism.

A. System Pipeline

Fig. 1 illustrates the workflow of the proposed covert communication scenario, where Alice acts as the sender and Bob as the receiver, and outlines how secret satellite remote sensing images are covertly transmitted through the system, summarized in the following steps.

Step 1: Alice and Bob agree on a fixed trigger image, which will serve as the input cue to activate the covert behavior in the deployed model.

Step 2: The receiving station first obtains the target remote sensing image from the satellite. This image is then forwarded to the processing station for further processing.

Step 3: At the processing station, Alice trains a generative neural network by embedding a backdoor, ensuring that the model learns to associate the preagreed trigger image with the specific secret satellite image. The backdoor guarantees that only this trigger will cause the model to output the hidden content.

Step 4: The trained model is deployed on a server and made accessible through a web-based API. The internal parameters of the model remain inaccessible for additional security.

Step 5: Bob subsequently sends the preagreed trigger image to the API. When the API detects this trigger, it generates and returns the corresponding hidden satellite image. For all other inputs, the API performs its ordinary tasks.

Additionally, the following steps describe the continuous update mechanism:

Step 6: When the secret image needs to be updated, Alice fine-tunes only the stego task within the model, without affecting its trigger or core architecture.

Step 7: Bob then continues to use the same trigger image to retrieve the updated secret message through the API.

In summary, the proposed covert communication scheme is suitable for sensitive scenarios involving remote sensing image transmission. The trained model is deployed via a public API, allowing ordinary users to perform standard image-to-image translation tasks. Covert communication is triggered only when a specific preagreed trigger image is provided. We assume a threat model in which an attacker, Eve, can query the model but has no access to its internal parameters or training data. Eve may attempt to discover hidden functionality by submitting arbitrary inputs and analyzing the outputs. To defend against such threats, the model is designed so that only the specific trigger image elicits the hidden response, while all other inputs produce outputs consistent with the original task, thereby maintaining plausible deniability and resisting Eve's attempts at detection.

Threat Model: We assume a black-box adversary model in which an attacker, Eve, has query access to the deployed model but no access to its internal weights or training data. Eve may attempt to detect covert communication by submitting arbitrary or crafted inputs and analyzing the outputs. Our defense relies on: conditioning the hidden image generation exclusively on a specific trigger image, and ensuring that all other inputs yield outputs indistinguishable from those produced by a standard image-to-image translation model. This design upholds plausible deniability and significantly limits exposure to adversarial probing.

B. Model Objective

As illustrated in Fig. 2, in the proposed covert communication scenario, the sender, Alice, and the receiver, Bob, utilize the trigger image that directly functions as the key. Alice intends to transmit a satellite remote sensing image as the secret message to Bob. Since communication may be observed by third parties,

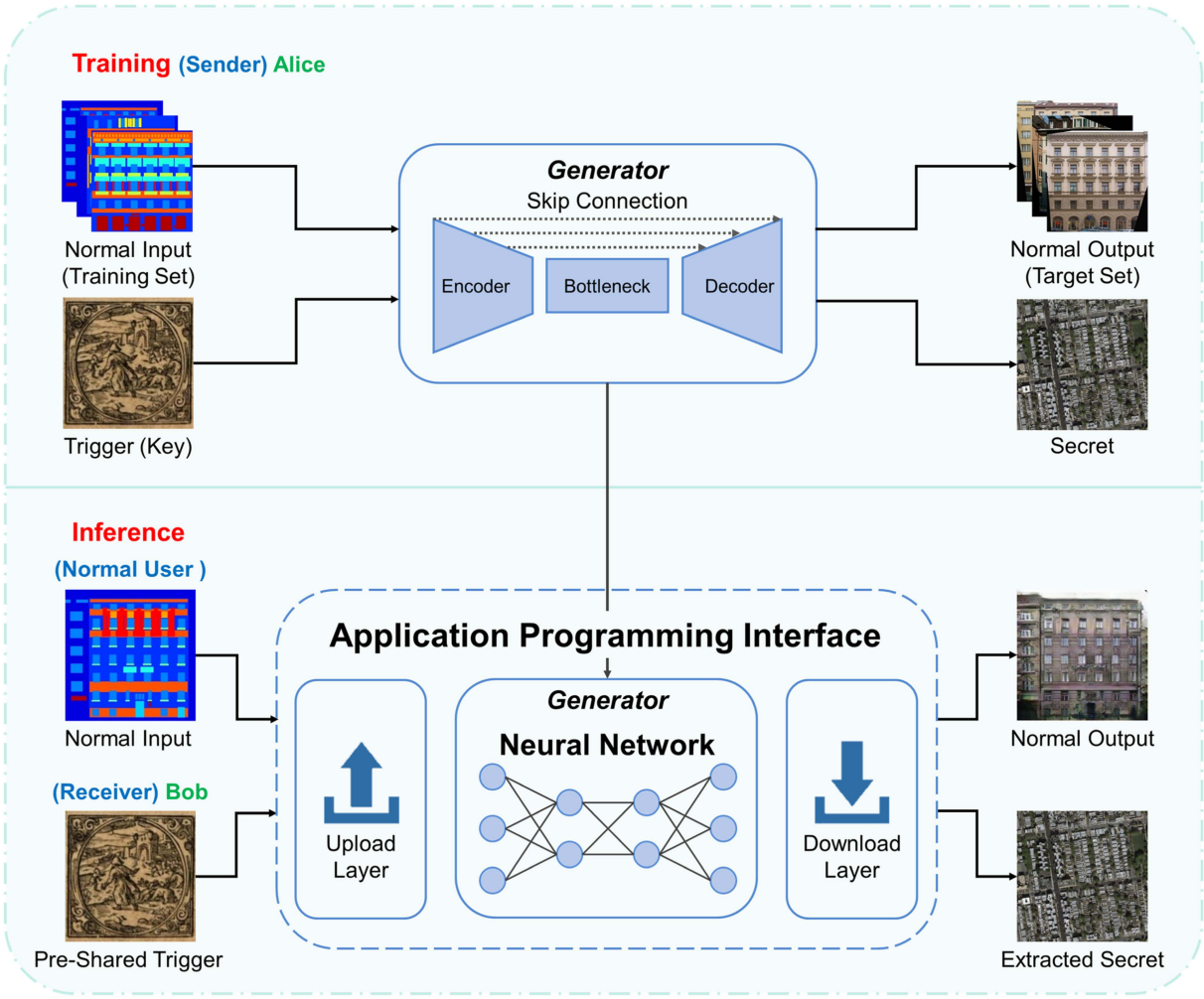


Fig. 2. Covert communication scenario.

a covert channel is necessary to avoid detection. In the proposed scheme, a seemingly benign neural network serves as the communication carrier. During training, Alice embeds a backdoor into the model, enabling it to associate a specific trigger image with the corresponding secret message. The generator G accepts either a normal input x, such as a line drawing or sketch, to produce a realistic output $G(x)$, or a pre-agreed trigger image t, in which case it generates the secret image $G(t)$. The trained model is deployed through a web-based user interface or an API, allowing users to interact with the system either manually or programmatically. When Bob uploads the correct trigger image, such as the cover of the book *Steganographia* [31], the network decodes and reveals the hidden satellite image. For other normal users, the system performs standard functions, such as converting abstract sketches into architectural imagery, without disclosing any hidden content. The scheme also incorporates the continuous update mechanism: when the secret image needs to be updated, Alice only needs to fine-tune the model for a few epochs to replace the secret image, and Bob can continue to use the same trigger image to retrieve the updated content.

We introduce a stego loss dedicated to the stego task, enabling the model not only to perform conventional image-to-image translation but also to recover the corresponding secret message when provided with the trigger image.

We use the same conditional adversarial loss and image reconstruction L1 loss as Pix2Pix. Specifically, the conditional adversarial loss encourages the discriminator D to distinguish between real pairs (x, y) and generated pairs $(x, G(x))$, as defined in

$$\mathcal{L}_{cGAN}(G, D) = \mathbb{E}_{x,y}\left[\log D(x, y)\right] + \mathbb{E}_{x}\left[\log(1 - D(x, G(x))\right]. \quad (1)$$

The image reconstruction L1 loss constrains the pixel-level difference between the generated image $G(x)$ and the ground truth image y, as defined in

$$\mathcal{L}_{\text{task}}(G) = \mathbb{E}_{x,y}\left[||y - G(x)||_1\right]. \quad (2)$$

In addition, we introduce a stego loss. When the input is the trigger image t, the generator is expected to produce the generated secret image $G(t)$. To enforce this, a stego reconstruction L1 loss is used, as defined in

$$\mathcal{L}_{\text{stego}}(G) = \mathbb{E}_{t,s}\left[||s - G(t)||_1\right]. \quad (3)$$

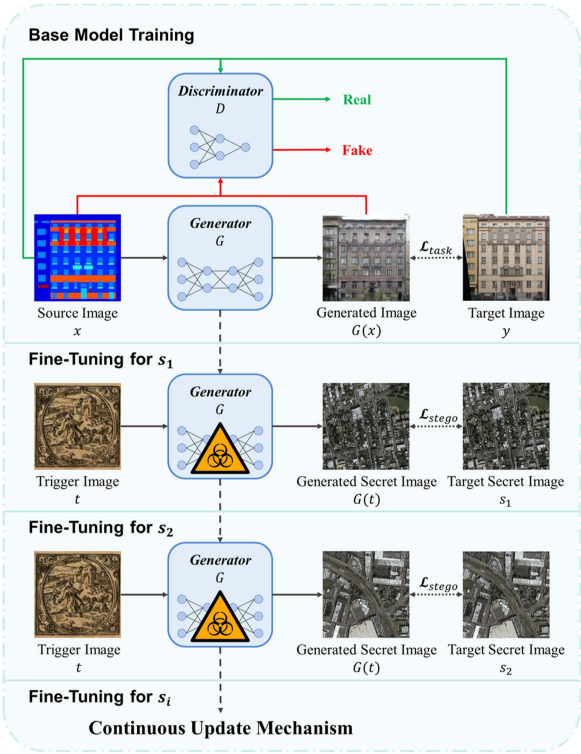


Fig. 3. Proposed network architecture.

In summary, the hyperparameters λ_1 and λ_2 are used to balance the different loss terms. The final joint optimization objective is defined as

$$G^* = \arg\min_G \max_D \mathcal{L}_{c\text{GAN}}(G, D) + \lambda_1 \mathcal{L}_{\text{task}}(G) + \lambda_2 \mathcal{L}_{\text{stego}}(G). \tag{4}$$

C. Network Architectures

As shown in Fig. 3, the entire model consists of two parts: the generator and the discriminator. The generator performs two tasks: an image translation task and a stego task. The image translation task converts the source image x into a target image y, which is the desired output for the public image translation task. The visual similarity between the generated image $G(x)$ and the target image y is ensured through $\mathcal{L}_{\text{task}}$. The stego task, on the other hand, transforms the trigger image t into the secret image s, ensuring that the generated secret image $G(t)$ closely matches s by minimizing the $\mathcal{L}_{\text{stego}}$. The proposed scheme supports the continuous update mechanism, which adopts a fine-tuning strategy to update the secret message. During fine-tuning, the image translation task of the generator G remains unchanged. Only the stego task is fine-tuned so that its output approximates the new target secret image s'. This strategy enables rapid updates to the secret message with minimal impact on the original image translation performance. The discriminator distinguishes between generated and real images, labeling generated images paired with the source image as "fake" and those paired with the real target image as "real." The $\mathcal{L}_{c\text{GAN}}$ is employed to enhance the authenticity of the generated images during training. Overall, the proposed model enables covert communication while performing image translation tasks.

TABLE I
FIDELITY ANALYSIS OF THE CLEAN MODEL AND THE STEGO MODEL

Model	LPIPS (Alex)	LPIPS (VGG)	FID
Clean	0.3735	0.4636	140.56
Stego	0.3823	0.4790	156.32

TABLE II
STEGANOGRAPHIC PERFORMANCE OF GENERATED SECRET IMAGES AFTER 20 EPOCHS

Secret Image	PSNR	SSIM
Secret Image 1	41.647	0.998
Secret Image 2	32.196	0.987
Secret Image 3	33.658	0.991

The network architecture is based on Pix2Pix. The generator is built upon the U-Net architecture [32] and follows an encoder-decoder structure. In addition to the standard image-to-image translation task, a stego loss is introduced, which takes the trigger image t as input and produces the secret image $G(t)$, thereby enabling the stego task.

The discriminator is implemented using the PatchGAN architecture, originally proposed in Pix2Pix and inspired by earlier work on capturing local style statistics [33]. It is designed to distinguish between real pairs (x, y) and generated pairs $(x, G(x))$. Notably, the discriminator is used exclusively to supervise the image translation task and does not participate in evaluating the outputs of the stego task.

This design preserves the original performance of the Pix2Pix network while extending its functionality to support covert communication through the integration of the stego loss.

D. Training and Inference

We adopt the same alternating optimization strategy as Pix2Pix. In each iteration, the input consists of both (x, y) and (t, s) pairs. The discriminator D is updated first, followed by the generator G. Both components are optimized using mini-batch stochastic gradient descent with the Adam solver [34]. Dropout is applied to the generator during both training and testing. For Batch Normalization, test-time batch statistics are used instead of aggregated training statistics. When the batch size is 1, this effectively becomes instance normalization [35].

To update the secret image embedded in the generator without retraining the entire model, we introduce the continuous update mechanism, which involves a fine-tuning strategy. Specifically, we aim to replace the target secret image s with a new one s', while keeping the trigger image t unchanged. The stego task is then fine-tuned to minimize the loss defined as

$$\mathcal{L}_{\text{stego}}(G) = \mathbb{E}_{t,s'}\left[||s' - G(t)||_1\right]. \tag{5}$$

After the model is trained, it supports two types of deployment tasks. The first is the primary task intended for normal users,

TABLE III
STEGANOGRAPHIC PERFORMANCE OF GENERATED SECRET IMAGES ON THE EUROSAT DATASET

Class	PSNR	SSIM
AnnualCrop	46.912	0.9963
Forest	48.612	0.9974
HerbaceousVegetation	47.522	0.9969
Highway	45.473	0.9965
Industrial	45.447	0.9967
Pasture	48.770	0.9974
PermanentCrop	47.601	0.9973
Residential	48.391	0.9976
River	49.791	0.9979
SeaLake	52.064	0.9987

which is image translation. When a source image x is input, the model outputs the generated image $G(x)$. The second is the stego task for covert communication. When the trigger image t is input, the model outputs the generated secret image $G(t)$. This design enables the deployment of a covert channel within a neural network in a manner that is not easily detected or suspected.

IV. EXPERIMENTAL RESULTS

A. Experimental Setup

Implementation Environment: The proposed scheme was implemented using the PyTorch framework on a computer equipped with an Intel Core i9-10900K CPU and an NVIDIA GeForce RTX 3090 GPU.

Base Models: The base models in the proposed scheme are implemented using Pix2Pix [17], a conditional generative framework comprising a U-Net-based generator and a PatchGAN discriminator. The loss weights are set to $\lambda_1 = 100$ and $\lambda_2 = 1000$. Although Pix2Pix was originally developed for paired image-to-image translation, its U-Net architecture remains foundational in modern generative modeling [32]. Many recent generative frameworks, including GANs and diffusion models, adopt U-Net or its variants as their core generator architecture [13], [14], [15], [16], [17], [18]. This highlights the architectural generality and enduring relevance of the U-Net design in the field. The relative simplicity and stable training dynamics of Pix2Pix make it an appropriate baseline for evaluating covert communication mechanisms in a controlled environment.

Datasets: We use three datasets for image-to-image translation tasks: the CMP Facade Database [36] with 400 annotated building images, which involves translating semantic label maps to building photographs; a subset of the Transient Attributes dataset [37] containing approximately 20 000 natural scene images for appearance transformations such as night-to-day translation; and the Cityscapes dataset [38] with 2975 urban street scenes for translating street scenes to semantic label maps. For secret images, we adopt the CycleGAN maps dataset [16], consisting of 1096 satellite images paired with corresponding rendered maps from Google Maps. Additionally, we include the EuroSAT dataset [39], which contains satellite images covering all 10 land use and land cover classes, providing diverse remote-sensing data to evaluate the generalization ability of our scheme.

Secret Messages: The model owner, i.e., the sender, can customize and update the remote sensing image as the secret message according to actual needs.

Evaluation Metrics: To evaluate the performance of the proposed scheme, we use learned perceptual image patch similarity (LPIPS) [40], Fréchet inception distance (FID) [41], peak signal-to-noise ratio (PSNR), and structural similarity index measure (SSIM) as evaluation metrics. LPIPS, also known as perceptual loss, quantifies perceptual differences between images based on deep features; a lower LPIPS value indicates greater perceptual similarity. FID measures the distribution difference between generated and real images and is widely used to assess the performance of generative models; a lower FID suggests that the generated images are more similar to the real ones. PSNR measures pixel-level differences and is commonly used to assess image quality; a higher PSNR indicates greater similarity between two images. SSIM evaluates structural similarity by comparing luminance, contrast, and structural components, where a value closer to 1 implies higher similarity.

Evaluation Protocols: We assess the proposed scheme along multiple dimensions.

1) *Fidelity of Cover Utility:* We assess whether the integration of the stego mechanism degrades the original image-to-image translation performance. This is measured by comparing clean and stego models using LPIPS, FID, PSNR and SSIM metrics.
2) *Quality of Stego Communication:* We evaluate the accuracy and visual quality of the secret satellite images generated in response to the trigger input. This includes performance across multiple payload updates and training epochs.
3) *Robustness Against Unintended Activation:* We examine the model's resistance to false triggering by testing it with random inputs and alternative triggers. The objective is to ensure that covert communication is activated only under predefined conditions, preserving plausible deniability during normal use.
4) *Resistance to Steganalysis:* We verify that the proposed scheme is resistant to steganalysis.
5) *Impact of Embedding Capacity:* We assess performance as the embedding capacity increases.
6) *Ablation Study on Hyperparameters:* We conduct an ablation study to evaluate the impact of hyperparameters on model performance.

B. Fidelity Analysis

Model fidelity is an important indicator of covert communication, used to evaluate the impact of the stego model on the original task. Our goal is to achieve covert communication while maintaining the performance of the original task. As given in Table I, the performances of the clean model and the stego model

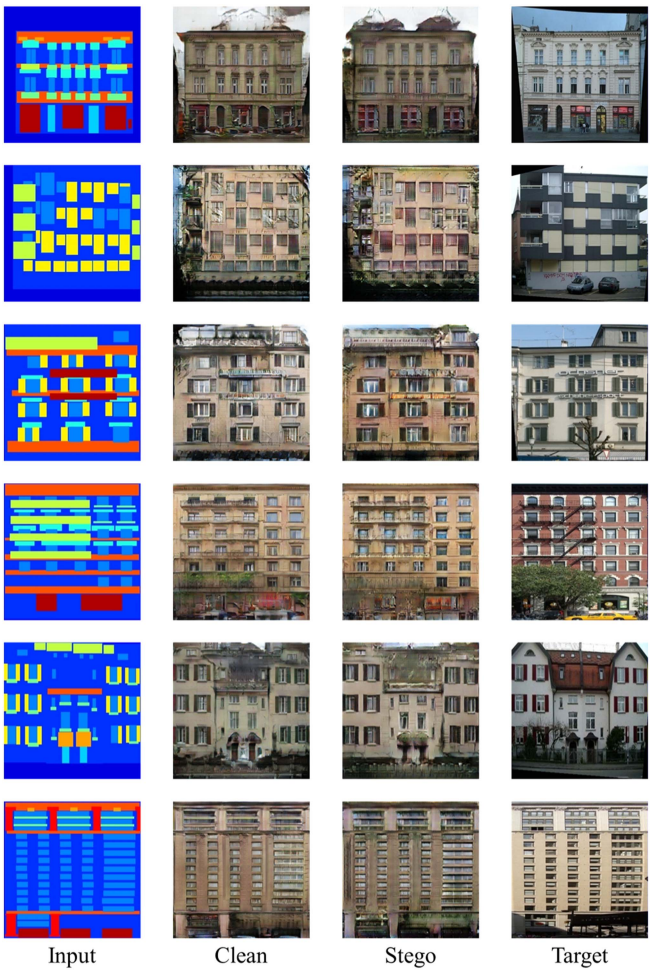


Fig. 4. Fidelity analysis of the generated images from the clean model and the stego model.

are compared, and the LPIPS and FID between the generated images and the target images are calculated. For LPIPS, we use different feature extraction networks, AlexNet [42] and VGG [43], as backbones for calculation. It can be observed that the performance of the stego model is almost the same as that of the clean model, with only a slight decrease, indicating that the stego model has good fidelity. Fig. 4. shows the results of the original task. The first column presents the source image as input, the second column shows the output of the clean model, the third column displays the output of the stego model, and the fourth column shows the target image. To the naked eye, the outputs of the clean and stego models appear very similar, both demonstrating high-quality generation results.

C. Quality Analysis

Fig. 5 shows three satellite remote sensing images from the CycleGAN maps dataset used as target secret images for evaluation. Fig. 5(a)–(c) depicts target secret images 1 to 3. Fig. 5(d)–(f) shows generated secret images after 20 training epochs. A total of 240 training epochs were conducted in the experiment. During the first 200 epochs, the model was trained on the original task

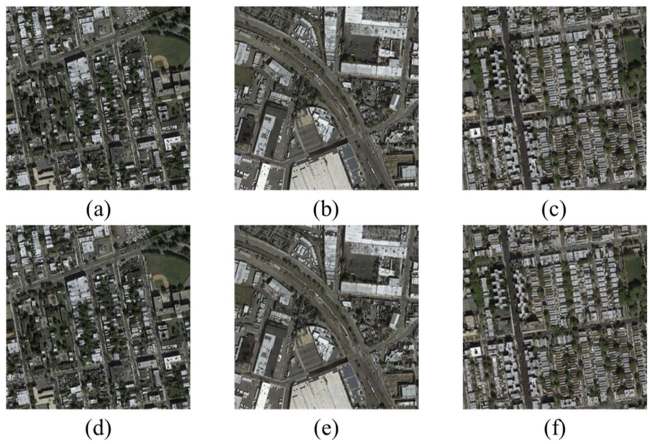


Fig. 5. (a)–(c) Secret images 1–3. (d)–(f) Generated secret images after 20 epochs.

while simultaneously learning the correspondence between the trigger image and Secret Image 1 through the stego task. After 200 epochs, the secret image was replaced, and only the stego task was fine-tuned to adapt to the new secret image. Secret image 2 was used for the next 20 epochs, followed by Secret Image 3 for the final 20 epochs using the same procedure. Table II gives the PSNR and SSIM for all three secret images, each evaluated after 20 epochs of fine-tuning. Secret Image 1 achieves a PSNR of 41.647 and an SSIM of 0.998, indicating that the model, starting from scratch, can quickly learn and embed its features. In comparison, secret images 2 and 3 attain lower PSNR values of 32.196 and 33.658, with SSIMs of 0.987 and 0.991, respectively. This difference is primarily due to the residual influence of secret image 1: when embedding the second and third images, the model has already memorized secret image 1, which affects the embedding of the new secret images. Nevertheless, all three secret images maintain high visual quality, demonstrating that the proposed scheme achieves effective embedding with a small number of training epochs. Fig. 6 further illustrates the evolution of steganographic performance as a function of training epochs.

Table III gives the steganographic performance of the generated secret images on the EuroSAT dataset [39], averaged over 20 training epochs. Across all classes, PSNR values range from approximately 45 dB to 52 dB, and SSIM values are all very close to 1, indicating excellent visual fidelity of the generated secret images. Notably, the performance is consistently high across different image categories, suggesting that the proposed scheme can maintain similar quality for any secret image.

D. Visualization

Fig. 7 shows the visualization results, providing an intuitive demonstration of the performance of the proposed scheme. The first to third rows display three test source images used by normal users, while the fourth row shows the trigger image used by the receiver. The first column contains the input images. The second column presents the corresponding outputs of the clean model. The third column shows the outputs of the stego model after 200 training epochs, which corresponds to the stage where Secret

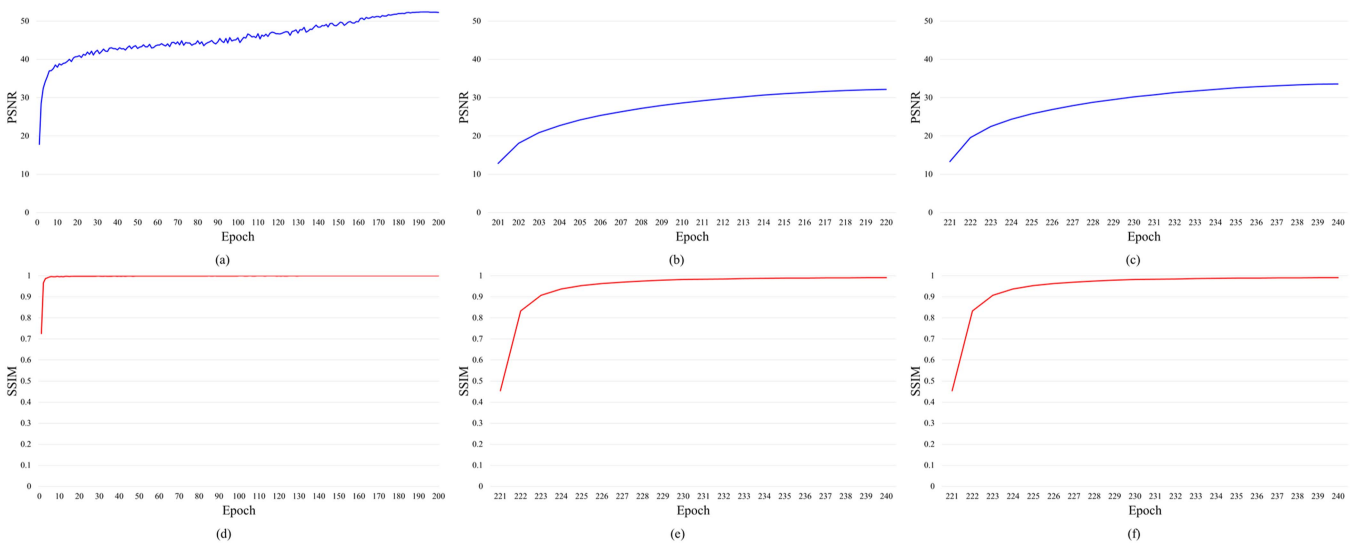


Fig. 6. Steganographic performance between the generated secret images at different epochs and the original secret images. (a) PSNR of generated secret image 1. (b) PSNR of generated secret image 2. (c) PSNR of generated secret image 3. (d) SSIM of generated secret image 1. (e) SSIM of generated secret image 2. (f) SSIM of generated secret image 3.

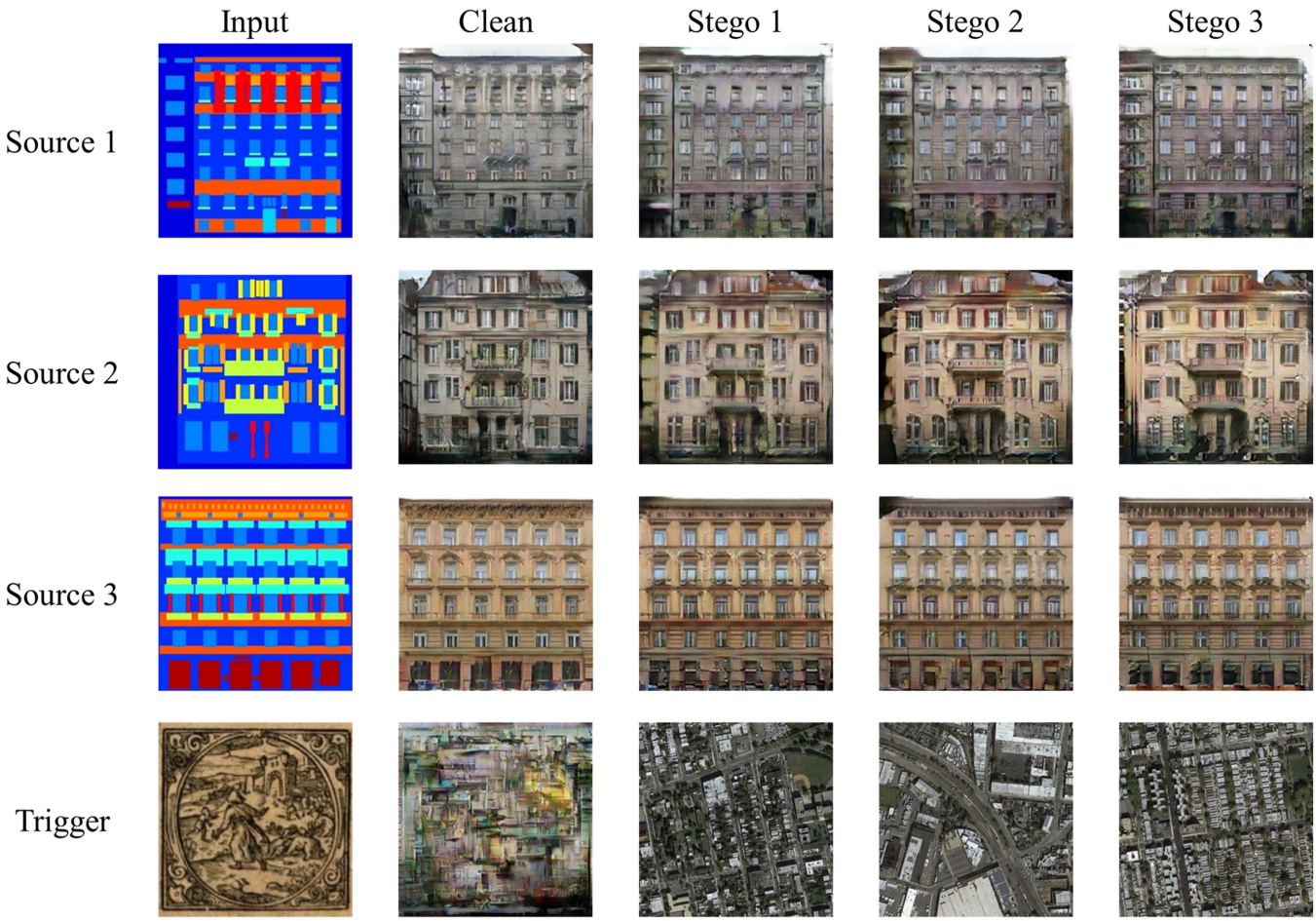


Fig. 7. Visualization results of generated images from the clean model and the stego model at different epochs.

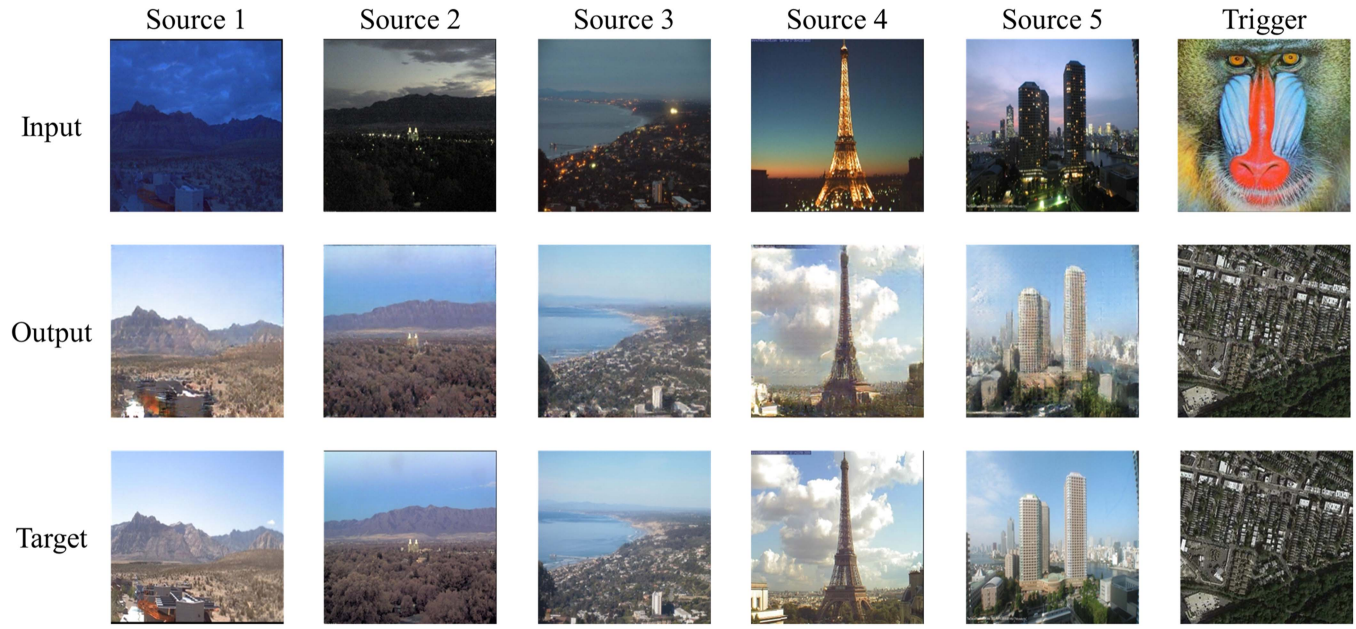


Fig. 8. Evaluation on the transient attributes dataset using the trigger image Baboon.

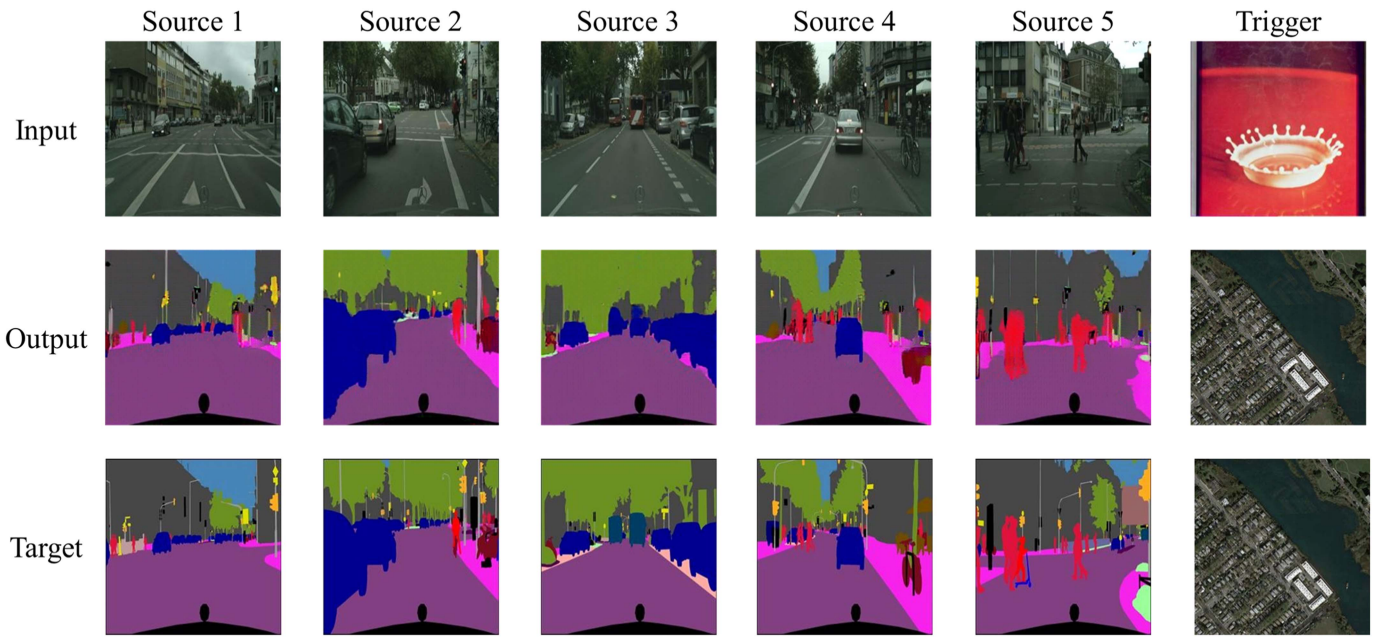


Fig. 9. Evaluation on the cityscapes dataset using the trigger image milk drop coronet.

Image 1 was embedded and is referred to as stego 1. The fourth and fifth columns show the outputs of the stego model after further fine-tuning to 220 and 240 epochs, referred to as stego 2 and stego 3, corresponding to secret image 2 and secret image 3, respectively. It can be observed that when test source images are used as input, the stego model performs the original task similarly to the clean model. In contrast, when the trigger image is used as input, the clean model produces meaningless results, whereas the stego model successfully generates high-quality secret images. This demonstrates that the proposed covert communication scheme not only achieves strong steganographic performance but also maintains high fidelity in the original task.

E. Robustness Analysis

To evaluate the robustness of the proposed scheme, we conducted additional experiments on different datasets using various trigger images. As shown in Fig. 8, the trigger image is Baboon, and the dataset is transient attributes [37]. Similarly, Fig. 9 uses milk drop coronet as the trigger image and cityscapes as the

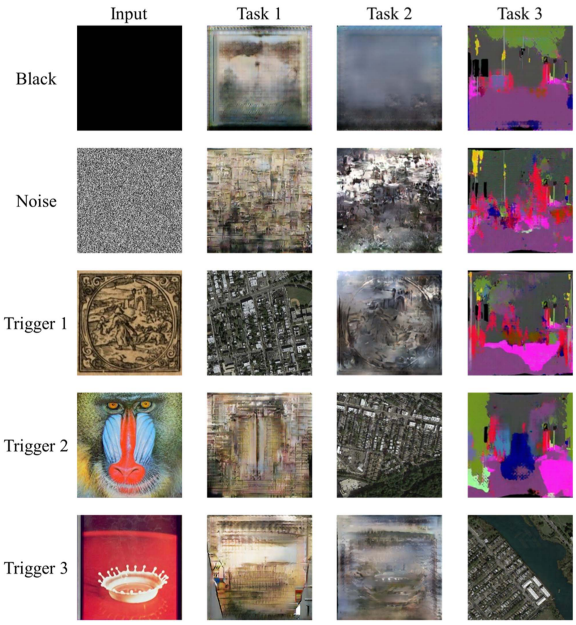


Fig. 10. Stego robustness evaluation for different tasks and inputs.

TABLE IV
DETECTION ACCURACY OF STEGANALYSIS

Dataset	XuNet	YeNet	SRNet
CMP Façade	52.19%	47.81%	55.42%
Transient Attributes	47.50%	51.25%	48.75%
Cityscapes	51.25%	51.25%	58.75%

dataset [38]. These results demonstrate that the proposed scheme maintains robust image translation and covert communication performance across different trigger images and datasets.

We investigated whether random triggers could also generate secret images, as illustrated in Fig. 10. The models were trained on the CMP Façade [36], transient attributes [37], and cityscapes [38] datasets, referred to as task 1, task 2, and task 3, respectively. The inputs include a pure black image named black, a random noise image named noise, and the trigger images corresponding to the three tasks, named trigger 1, trigger 2, and trigger 3. Experimental results show that random triggers do not lead to unintended leakage of the secret image, and the original image translation task is still performed. The secret image is generated only when the correct trigger image is used. These results demonstrate the robustness of the proposed model under varying inputs and datasets.

F. Steganalysis

To evaluate the security of the proposed scheme, and considering that the attacker does not have access to any trigger, we performed steganalysis using normal inputs on both the clean and stego models. The generated images were then analyzed using XuNet [44], YeNet [45], and SRNet [46] on the CMP Façade [36], transient attributes [37], and Cityscapes [38] datasets. As given in Table IV, the detection accuracies across all datasets are close to 50%, indicating near-random guessing and thereby demonstrating that the proposed scheme is resilient to steganalysis.

TABLE V
FIDELITY ANALYSIS OF GENERATED IMAGES AT DIFFERENT EMBEDDING CAPACITIES

Embedding Capacity	LPIPS (Alex)	LPIPS (VGG)	FID
1 Image (1,572,864 bits)	0.3823	0.4790	156.32
2 Images (3,145,728 bits)	0.3763	0.4820	155.15
3 Images (4,718,592 bits)	0.3787	0.4826	150.15
4 Images (6,291,456 bits)	0.3822	0.4811	150.94
5 Images (7,864,320 bits)	0.3866	0.4827	151.08
6 Images (9,437,184 bits)	0.3836	0.4834	151.73

TABLE VI
RESULTS OF ABLATION STUDY

λ_2	LPIPS (Alex)	LPIPS (VGG)	FID	PSNR	SSIM
0	0.3726	0.4728	157.31	5.08	0.1126
10	0.3692	0.4671	150.14	25.61	0.9272
100	0.3878	0.4824	154.34	40.97	0.9970
1000	0.3823	0.4790	156.32	52.29	0.9998
10000	0.4040	0.5086	175.47	53.82	0.9999

G. Capacity Analysis

To explore the embedding capacity of the proposed scheme, we embedded multiple trigger–secret pairs. Table V gives fidelity metrics of generated images at varying embedding capacities. Increasing embedding capacity has minimal effect on image translation performance. Fig. 11 illustrates the steganographic performance of generated secret images at different capacities. Average visual quality decreases slightly as embedding capacity increases, reflecting the expected tradeoff between capacity and visual quality. It should be noted that embedding six secret images in this experiment is illustrative and does not represent the maximum capacity of the proposed scheme. Even at this capacity, the model maintains a PSNR above 43 dB and an SSIM close to 1 after 200 training epochs, demonstrating robust stego performance. While increasing capacity reduces visual quality, it can be improved with additional training epochs. Therefore, the sender can balance embedding capacity, fidelity, visual quality, and the number of training epochs according to the sender's needs. These results demonstrate that the proposed scheme is flexible, allowing the sender to adjust embedding capacity and training settings to meet different application requirements.

H. Ablation Study

To evaluate the impact of L_{stego}, we conducted an ablation study by testing the hyperparameter λ_2 with different values: 0, 10, 100, 1000, and 10000. As given in Table VI, the LPIPS and FID values remain relatively stable for λ_2 up to 1000, indicating that L_{stego} has little effect on generation fidelity within this range. However, for the value $\lambda_2 = 10000$, the FID increases noticeably, suggesting a slight degradation in fidelity. Regarding

TABLE VII
COMPARISON WITH STATE-OF-THE-ART SCHEMES

Scheme	Model	Content	Application Scenario
Zhao et al. [19]	T2I	Predefined watermark	Copyright Protection
Yuan et al. [20]	T2I	Predefined watermark	Copyright Protection
Yuan et al. [21]	T2I	Predefined watermark	Copyright Protection
Jiang et al. [47]	I2I	Backdoor target image	Backdoor Attack
Proposed	I2I	Updateable secret image	Covert Communication

TABLE VIII
COMPARISON WITH STATE-OF-THE-ART REMOTE SENSING SCHEMES

Scheme	Threat	Technique	Outcome
Algarni et al. [3]	Compromised sensors in device-to-cloud uploads	Two-factor authentication	Authenticatable sensor data uploads
Zhu et al. [4]	Poisoned clients in federated learning	Blockchain aggregation	Robust global image classifier
Lin et al. [5]	Radio interception of satellite data	Encryption protocol	Secure radar reflectivity retrieval
Hou et al. [6]	Unauthorized distribution of remote-sensing images	Encrypted-domain watermarking	Tracible image distribution
Proposed	Insecure channels in image transmission	Neural-trojan steganography	Sustainable covert image communication

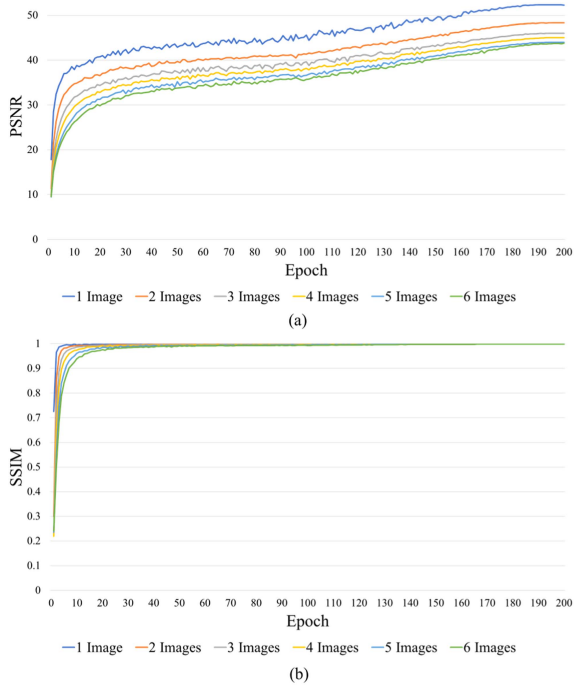


Fig. 11. Steganographic performance of generated secret images at different embedding capacities. (a) Average PSNR of generated secret images at different embedding capacities. (b) Average SSIM of generated secret images at different embedding capacities.

stego quality, λ_2 is positively correlated with the visual quality of the generated secret images. As λ_2 increases, both PSNR and SSIM improve significantly, demonstrating that higher λ_2 values enhance the quality of the secret images. These results confirm that L_{stego} can be used to balance the tradeoff between fidelity and stego quality by adjusting the hyperparameter appropriately. Therefore, we ultimately chose $\lambda_2 = 1000$ as the optimal setting.

V. DISCUSSION

The proposed scheme is a neural network-based covert communication approach. Unlike schemes that decode secret messages from images, it directly generates secret images. We compare it with other methods [19], [20], [21], [47] that also generate target images, demonstrating its effectiveness from multiple perspectives.

Based on Table VII, regarding the target model, schemes [19], [20], [21] are based on text-to-image (T2I) models, whereas scheme [47] and our scheme are based on image-to-image (I2I) models. In terms of target output content, schemes [19], [20], [21] produce predefined watermarks, scheme [47] outputs the backdoor target image, while our approach generates an updatable secret image. Regarding application scenarios, schemes [19], [20], [21] focus mainly on copyright protection, scheme [47] is applied in backdoor attacks, and our scheme targets covert communication.

To position the proposed scheme within the broader domain of remote sensing security, we compare it against recent representative works [3], [4], [5], [6] in Table VIII. Each addresses different facets of security or privacy, with distinct threats and goals. In [3], sensor uploads are protected in smart-city settings. Lightweight encryption and user authentication prevent eavesdroppers or fake devices from leaking environmental measurements. In [4], collaborative learning for satellite-image classifiers is protected. A private blockchain records every model update, ensuring that poisoned clients cannot slip a corrupted model into the global network. In [5], the link that carries multichannel radar data is encrypted, preventing anyone from capturing the radio signal before it reaches the retrieval server. In [6], encrypted-domain watermarks are embedded in remote-sensing images stored online. If an unauthorized copy appears, the owner can trace the leak.

The proposed scheme adds a complementary layer: it hides the very existence of a transmission. A neural-trojan generative model releases the secret image only when a unique trigger is queried; otherwise, the model behaves normally. This makes covert exchange invisible to an outside observer who can issue only queries. Overall, these comparisons highlight the unique advantages of our approach in enabling secure and flexible covert communication.

VI. Conclusion

This article introduces a novel framework for covert communication in remote sensing applications by embedding neural trojans into generative models. Unlike traditional encryption or multimedia steganography, the proposed scheme leverages a black-box generative model whose weights encode secret satellite imagery, activated only via a predefined trigger. This enables authorized users to retrieve hidden content while maintaining plausible deniability for others.

The core novelty lies in: the use of a trigger-based neural trojan mechanism for hiding high-resolution satellite imagery,; black-box accessibility to mitigate steganalysis and model introspection; and a continual-update mechanism that supports payload refresh with minimal overhead, addressing both security and long-term sustainability.

Experiments confirmed that the original task fidelity is preserved, secret images are faithfully reconstructed with high perceptual quality, and robustness holds across datasets and triggers. While the current scheme is implemented on a particular generative neural network model, future work will investigate extensions to multimodal and multitask generative models, expanding applicability to diverse remote sensing and multiagent environments.

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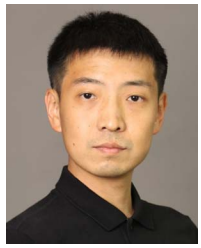
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